

CubeDesign 2026

CubeSat category

1. General Objective of the Competition

CubeDesign 2026 aims for participating teams to develop, document, test, and validate a CubeSat-based system capable of executing an ADS-B signal reception mission. The competition includes design, integration, operation, data analysis, and environmental testing stages.

2. Reglamentación General

- Each team must consist of **2 to 10 students** and **1 faculty advisor**.
- Schools, universities, institutes, or interdisciplinary groups may participate.
- Building a physical CubeSat is not mandatory. The system must comply with the dimensional requirements of the CubeSat Design Specification (CDS v14.1), and equivalent systems must perform the specified missions. Tests conducted during the competition are representative and do not guarantee launch qualification. Activities are organized by complexity, not necessarily chronologically.
- The first stage is a documentation phase. Each team must submit, via email, a complete project data package 2 months before the on-site tests. Only selected teams will advance to the second stage at UNSAM.
- Documentation must be submitted in English.

3. Competition Stages

The competition has two main stages:

3.1 Stage 1 – Documentation Evaluation

Two months before the on-site tests, each team must submit a **Design Package (DP)** including:

- Mission concept
- System architecture
- Mass, power, data, and financial budgets
- ADS-B payload design
- V&V plan
- Cost estimation and justification
- Complete engineering model description:
 - System block diagram
 - Communication scheme
 - Electrical architecture and power management
 - Implemented algorithms
 - Testing and validation procedures

Only selected teams will proceed to the on-site stage.

3.2 Stage 2 – On-Site Testing

To be held at UNSAM, including:

- Mission execution
- Basic, intermediate, and advanced tests
- Environmental testing (optional, with multiplier effect)
- Final presentation before the jury

4. Technical Tests

The tests are organized into levels of increasing complexity.

4.1 Basic Level

- **Communication test:** The CubeSat must receive telecommands (TC) [0.5 points wired, 1 point RF – TBC] and send telemetry (TM) [0.5 points wired, 1 point RF – TBC].

4.2 Intermediate Level

- **Mechanism test:** The CubeSat must deploy an antenna or similar device via remote command. The deployed element must be at least 10 cm. [0/1 point – TBC]. Teams must bring their own ground station.
- **Battery charging using a light source:** current and voltage will be evaluated. [0/1 point – TBC].

4.3 Advanced Level

- **Detumbling:** reduce initial rotation (60 rpm TBC) to below 5 rpm (TBC). Target times: <10 s, <25 s, <60 s.
- **Attitude determination (azimuth):** measurement based on a light source. [$\leq 5^\circ$, $\leq 15^\circ$, $\leq 30^\circ$]

4.4 Mission 2026: ADS-B Mission for Air Surveillance in Remote Areas

The increasing air traffic and the need for surveillance in regions without adequate ground coverage represent a realistic and relevant technical challenge. Satellite-based ADS-B systems are an emerging solution for aircraft detection, tracking, and trajectory reconstruction in oceans, deserts, mountain ranges, and polar regions.

The CubeDesign ADS-B mission requires each team to design, integrate, and validate a CubeSat-like system capable of **receiving, and processing ADS-B** messages from an SDR-based test environment.

The system must:

- Receive ADS-B signals at 1090 MHz
- Classify, store, and transmit them to the ground station
- Decode standard ADS-B messages and extract:
 - ICAO identifier
 - Position
 - Altitude
 - Velocity
- Handle multiple aircraft simultaneously

Teams must define whether processing is performed onboard, on the ground, or hybrid, and justify the choice. The mission must be conducted for **10 continuous minutes**, with a maximum of **20 aircraft per test**.

Additionally, each team **must propose a secondary mission compatible with the main mission**. Implementation is not mandatory.

4.4.1 Mission Objectives

Primary objective

Design and validate a CubeSat system capable of:

- Detecting ADS-B signals from aircraft
- Decoding position, velocity, heading, and altitude
- Reconstructing trajectories and estimating origin and destination
- Transmitting processed data as telemetry

Objetivos secundarios

- Evaluate system architecture decisions
- Analyze trade-offs between coverage, latency, and power
- Strengthen technical documentation practices

4.4.2 System Architecture

Each team must implement a system including at least:

- CubeSat platform (1U to 3U)
- OBC
- EPS with battery
- Communication system
- ADS-B payload
- ADS-B antenna (1090 MHz)
- Decoding and processing software
- Ground segment for control and analysis

4.4.3 ADS-B Payload

The payload must:

- Receive ADS-B signals
- Decode standard messages
- Extract at least:
 - ICAO address
 - Position (latitude, longitude)
 - Altitude
 - Velocity
- Generate structured telemetry

4.4.4 Test Environment

The organization will provide:

- Simulated ADS-B scenarios using SDR
- Multi-aircraft traffic

5. Software and Data Management

The system must include software capable of:

- Managing operational modes
- Executing the mission according to the defined ConOps

Mission and telemetry data must be:

- Structured
- Documented
- Reproducible
- Available for post-analysis

6. Verification and Validation (V&V)

Each team must define a V&V strategy demonstrating compliance with all high-level requirements.

The V&V plan must include:

- Verification methods (test, analysis, inspection, demonstration)
- Justification of methods
- Traceability between requirements and tests

Not all planned tests will be executed. Only those defined by the organization will be performed.

7. Communications Test

The CubeSat must:

- Receive telecommands and transmit telemetry
- Wired and RF modes will be scored differently

8. Environmental Testing (Optional – Multiplier Effect)

Only for teams passing Fit-Check inspection.

The system must execute the mission before testing. After thermal and vibration tests, it must:

- Demonstrate functional survival, or
- Justify limitations

based on telemetry analysis.

Thermal Tests

The system will undergo thermal cycling testing to evaluate its functional performance and the response of critical subsystems to temperature variations. Initially, two (2) thermal cycles are planned, with the following reference parameters:

- Range: -10°C to $+50^{\circ}\text{C}$ (TBC)
- Ramp rate: $\sim 1.5^{\circ}\text{C}/\text{min}$
- Stabilization: 30 minutes
- Battery must remain above 0°C

Note:

1. *If the system includes an internal battery, its temperature shall be maintained above 0°C at all times during the test. If necessary, the team must implement and justify active thermal control strategies to meet this requirement.*
2. *Alternatively, teams **may remove the internal battery during thermal vacuum testing and operate the system using an external power source**, provided that:*
 - a. *the system remains fully functional during the test, and*
 - b. *the selected configuration is clearly documented and justified.*
3. *The final parameters of the thermal test, as well as the number of cycles, may be adjusted depending on the number of participating teams, infrastructure availability, and total event duration. These parameters will be confirmed and communicated by the organization prior to testing.*

Vibration Test

Random vibration test in accordance with NASA/GEVS-14.1 requirements: 2 minutes per axis at 1 g.

- Environmental testing will take place after the mission is completed and is not mandatory. Survival results in a 1.4 multiplier on mission results.

- To conduct environmental testing, it is mandatory to pass a “Fit-Check” inspection (which verifies the dimensions of the mechanical enclosure specified in the CDS) and inspect loose parts (which could cause accidents during vibration testing).

The system must be capable of performing its mission prior to environmental testing. Following the vibration and thermal tests specified by the organization, the system must:

- demonstrate functional survival, or
- justify the observed limitations,

by analyzing the telemetry data collected before, during, and after the tests.

8.1 Thermal Tests

- Range: -10°C to $+50^{\circ}\text{C}$ (TBC)
- Ramp rate: $\sim 1.5^{\circ}\text{C}/\text{min}$
- Stabilization: 30 minutes
- Battery must remain above 0°C

8.2 Vibration Test

- Random profile per NASA GEVS
- 14.1 Grms, 2 minutes per axis
- Inside of the Test-POD

8.3 Post-testing

- Send any telemetry data to demonstrate subsequent operation.

9. Final Presentation

Each team will present:

- CubeSat design
- Test results
- Environmental telemetry
- Performance analysis

Evaluated by a specialist panel.

8. Evaluation System

Final score is based on:

- Documentation
- Technical tests
- Mission
- Environmental tests
- Final presentation

8.1 Budget & Fair Play

Teams must present:

- Total budget
- Subsystem costs
- COTS vs custom justification

An efficiency factor will consider:

- Performance
- Cost
- Technical coherence

8.2 Scoring System and Awards

All categories contribute to a single overall score, with defined weightings. Scoring details will be specified in the final version of the rules.

8.3 Special Awards (non-exclusive)

- *Best System Design*
Clear, coherent, and defensible architecture
- *Best ADS-B payload implementation*
Technical performance of the receiver and processing
- *Best technical documentation*
Clarity, traceability, and professional standard
- *Grand Prize Winner: CubeDesign CubeSat*
Highest weighted total score

9. General Notes

- The tests do not guarantee qualification for actual flight.
- Teams must bring their own ground station.
- Clarifications and supplementary materials will be published prior to the event.

High Level Requirements

CubeSat ADS-B Mission

1. General

HLR-GEN-01

The system shall conform to a mechanical envelope compatible with the CubeSat Design Specification (CDS v14.1), within a volume equivalent to 1U, 2U, or 3U.

Verification: Inspection (Fit-Check)

HLR-GEN-02

The system shall include, at minimum, the following subsystems:

- Structural platform
- Electrical Power System (EPS) including battery
- On-Board Computer (OBC)
- Communication subsystem
- Attitude Determination and Control System (ADCS, minimum functional capability)
- ADS-B payload
- Flight and mission software

Verification: Inspection + Documentation review

HLR-GEN-03

The system shall operate autonomously during mission execution, supporting remote interaction via telecommands (TC) and generating telemetry (TM) without manual intervention.

Verification: Documentation review

HLR-GEN-04

Each team shall define, document, and maintain a consistent set of subsystem-level requirements derived from system-level requirements.

Verification: Documentation review

2. Communications (TM/TC)

HLR-COMM-01

The system shall receive telecommands (TC) from the ground segment and execute corresponding actions within a bounded response time defined by the team.

Verification: Test

HLR-COMM-02

The system shall generate and transmit structured telemetry (TM), including at minimum:

- System status
- Power parameters
- Attitude information
- Mission data

Verification: Test + Data inspection

HLR-COMM-03

The communication architecture (including physical layer, protocols, and data rates) shall be defined, justified, and consistent with mission requirements.

Verification: Documentation review + Test

3. Power and Energy Requirements (EPS)

HLR-EPS-01

The system shall include a rechargeable energy storage element enabling autonomous operation for the full mission duration.

Verification: Test

HLR-EPS-02

The system shall demonstrate the capability to recharge its battery using a light-based energy source representative of solar input.

Verification: Test

HLR-EPS-03

The system shall define and justify a power budget covering all operational modes, including nominal and peak consumption.

Verification: Documentation review

HLR-EPS-04

The system shall maintain safe battery operating conditions (voltage, current, temperature) throughout all test phases.

Verification: Test + Telemetry analysis

HLR-EPS-05

The system shall support operation from an external power source during testing phases, including thermal vacuum (TVAC), without requiring modification of core subsystems.

If an internal battery is present, the system shall allow it to be removed during such tests while maintaining full system functionality.

The external power interface shall provide electrical characteristics equivalent to nominal operating conditions and shall not interfere with system performance.

Verification: Test + Inspection

4. Attitude Requirements (ADCS)

HLR-ADCS-01

The system shall estimate its attitude along at least one axis using onboard sensors.

Verification: Test

HLR-ADCS-02

The system shall reduce an initial rotational condition (detumbling) to below 5 rpm (TBC) within a finite and measurable time.

Verification: Test

HLR-ADCS-03

The system shall quantify and report attitude determination performance, including error metrics and limitations.

Verification: Analysis + Documentation review

5. ADS-B Mission Requirements

HLR-ADS-01

The system shall receive ADS-B signals at 1090 MHz and process them during mission execution.

Verification: Test

HLR-ADS-02

The system shall decode standard ADS-B messages and extract, at minimum:

- ICAO identifier
- Position (latitude, longitude)
- Altitude
- Velocity

Verification: Test + Data validation

HLR-ADS-03

The system shall detect and process multiple aircraft simultaneously within a single mission scenario.

Verification: Test

HLR-ADS-04

The system shall implement and justify a processing architecture (onboard, ground, or hybrid).

Verification: Documentation review

HLR-ADS-05

The system shall execute a continuous mission of 10 minutes, processing up to 20 aircraft per test.

Verification: Test + Data validation

HLR-ADS-06

Each team shall define a secondary mission concept compatible with the primary mission.

Verification: Documentation review

Note: This mission does not necessarily have to be implemented in hardware or software.

HLR-ADS-07

The system shall associate a timestamp with each received or processed ADS-B message, with a time resolution sufficient to enable trajectory reconstruction and temporal ordering of events within the mission duration.

Verification: Test + Data inspection

HLR-ADS-08 - Mission Success Criteria

The system shall meet defined mission success criteria during the 10-minute operational window, including at minimum:

- Successful reception of ADS-B signals
- Decoding of valid ADS-B messages
- Extraction of required parameters (ICAO, position, altitude, velocity)
- Transmission of processed data as telemetry

Each team shall define quantitative success metrics, including at least:

- Minimum number or percentage of correctly decoded messages
- Maximum acceptable data loss rate
- Maximum acceptable latency between reception and telemetry transmission

Verification: Test + Data inspection

6. Software and Data Requirements

HLR-SW-01

The system shall implement operational modes consistent with the defined Concept of Operations (ConOps).

Verification: Test + Documentation review

HLR-SW-02

Mission and telemetry data shall be:

- Structured
- Documented

- Reproducible

Verification: Data inspection

7. Verification and Validation Requirements

HLR-VV-01

Each team shall define a Verification and Validation (V&V) strategy demonstrating compliance with all system requirements.

Verification: Documentation review

HLR-VV-02

Verification methods (test, analysis, inspection, demonstration) shall be explicitly defined and justified.

Verification: Documentation review

8. Environmental Requirements

HLR-ENV-01

The system shall successfully execute the mission prior to environmental testing.

Verification: Test

HLR-ENV-02

The system shall demonstrate functional survival after thermal and vibration testing, or provide justified analysis of observed limitations.

Verification: Test + Analysis

9. Cost and Fair Play Requirements

HLR-COST-01

Each team shall provide a cost estimation for the complete system, including all subsystems and major components.

Verification: Documentation review

HLR-COST-02

Design decisions shall be justified based on performance, cost, and complexity trade-offs.

Verification: Documentation review

10. System Coherence Requirement

HLR-SYS-01

The mission definition, system architecture, requirements, verification plan, and results shall be consistent and defensible from a systems engineering perspective.

Verification: Integrated review